

Grace Chang

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WORK EXPERIENCE

Interface Software Engineer · *Microsoft (Experiences & Devices)* Nov 2024 – Present

- Led end-to-end interface design and implementation for the Windows Privacy Settings experience serving 4M+ monthly active devices, partnering closely with designers and PMs to translate CSAT telemetry analysis into UI refreshes and architectural decisions that lifted feature discoverability by 10% and cut render latency by 1.57s.
- Designed and built a scalable custom UI component library to replace fragmented implementations across 35+ complex, data-driven interfaces, cutting rendering time by 33% and establishing a consistent design system that raised craftsmanship across surfaces.
- Pioneered an LLM-powered developer experience platform that transformed engineering knowledge into interactive AI workflows through MCP tool orchestration, reducing new-hire onboarding friction by 87.5%.

Distributed Software Engineer · *Microsoft (Azure Core)* Aug 2023 – Nov 2024

- Optimized distributed infrastructure serving 40B+ daily requests, improving P99.9 latency by up to 250% through architectural improvements and synchronization optimization.
- Designed adaptive workload management for Azure Resource Manager supporting 20K VM admissions/minute and helping scale the service to 700K requests/hour through a staged, feature-flag-gated rollout across dogfood, canary, and worldwide traffic tiers.
- Isolated Cosmos DB containers by query pattern, cutting average CPU usage from 70% to 3% and reducing resource query duration from 330ms to 5ms.

PROTOTYPING & RESEARCH EXPERIENCE

Interactive Design Engineer · *MIT Media Lab, Tangible Media Group* Sep 2022 – May 2023

- Designed and prototyped [Sonic Stitches](#), a wearable telepresence system exploring haptic and tactile presence — engineering conductive-embroidery capacitive sensing on Arduino and mapping touch gestures to real-time generative audio-visual feedback, enabling remote users to feel each other's presence through a physical textile medium and exemplifying a new human-computer interaction paradigm.

Creative Technology Engineer · *MIT Media Lab, Personal Robots Group* Mar 2022 – Jun 2022

- Prototyped expressive interaction behaviors for the Jibo social robot — integrating character animation, haptic feedback, and conversational interaction — and partnered with designers and researchers to rapidly prototype and user-test expressive personas in short timeframes, iterating on interaction flows to raise engagement in robot-fostered parent-child reading sessions.

Interaction Prototyping Engineer · *MIT Media Lab, Fluid Interfaces Group* Oct 2021 – Mar 2022

- Prototyped and engineered [Expressive Navigation](#), an affect-aware iOS system using Apple's Vision framework (computer vision) and UIKit to infer emotional cues from facial expressions and deliver real-time feedback — applying perceptual and affective computing to help clinicians track patient emotional shifts across therapy sessions.

AR/VR Prototyping Engineer · *MIT Climate & Sustainability Consortium* Jun 2021 – Oct 2021

- Built a real-time driving simulation in Unity and extended it into VR via the Unity XR Device Simulator, rendering spatial acceleration and motion cues that conveyed an autonomous agent's braking, turning, and lane-change decisions to teach safe-driving strategies through immersive spatial interaction.

PROJECTS

[AI Temperature Monitor](#) — *ML-Driven Intelligent Sensing System*

- Engineered an AI-powered temperature monitoring system integrating Arduino sensors, TensorFlow forecasting, and automated control to demonstrate predictive modeling and intelligent system design for organization-wide technical training.

[Magdalene X²](#) — *Multi-Agent Health Intelligence iOS App*

- Architected and shipped an autonomous AI health companion iOS app to bridge fragmented healthcare experiences by unifying personal health signals across labs, cycle, nutrition, fitness, and sleep, leveraging multi-agent workflows and iterative product design to deliver personalized health insights.

[Dispatcher Voice Agent & Workflow Automation](#) — *Freight Dispatch & Carrier Network Platform*

- Built event-driven AI voice agents triggered by inbound calls and webhook events to automate carrier-to-load coordination for a network of ~2,000 active carriers moving ~500 loads/week, deployed via Docker to Azure Container Apps and launched end-to-end into production.

[Residential Property Servicing Scheduler](#) — *Automated Vendor Coordination Service*

- Diagnosed a production incident causing dropped webhooks, resolved it with backoff-and-jitter retry logic, and built a daily cron-driven pipeline automating a ~\$80K/month vendor-scheduling program with live calendar matching and failure-mode safeguards.

SKILLS

IOS & APPLE PLATFORMS	Swift, UIKit, Vision, iOS App Development
HCI & PROTOTYPING	Figma, Cursor, Unity, TouchDesigner, Arduino, Interaction Design, Physical Computing, Multimodal Interfaces
AI & INTELLIGENT SYSTEMS	LLMs, Agentic Systems, MCP / Tool Orchestration, Prompt Engineering, Tool Calling, TensorFlow
SOFTWARE ENGINEERING	Python, C++, C#, TypeScript, JavaScript, React, Vue, Node.js, Flask, Git, HTML, CSS
SYSTEMS	Azure, Distributed Systems, Docker, Kubernetes, Cosmos DB, MongoDB, MySQL

EDUCATION

[Bachelor of Arts in Computer Science \(Dual Enrollment\)](#) May 2023

Massachusetts Institute of Technology · Cambridge, MA

Tangible User Interface · Artificial Intelligence · Software Construction · Software Studio · Data Structures Algorithms

Wellesley College · Wellesley, MA